Computer Networks Manual

III Sem BCA

1. Execute the following commands:

a. **arp:** Arp stands for Address Resolution Protocol. It maps a known IP address to an unknown MAC address. It allows a router to encapsulate a packet correctly before forwarding.

Syntax: arp -a

```
C:\>arp -a

Interface: 192.168.29.32 --- 0x6
  Internet Address      Physical Address      Type
  192.168.29.1          8c-a3-99-42-77-a6     dynamic
  192.168.29.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
```

b. **ipconfig:** ipconfig stands for IP Configuration. It is a Windows command designed to display current TCP/IP network configurations but can also be used to modify your DNS and DHCP settings.

Syntax: ipconfig

```
C:\>ipconfig

Windows IP Configuration


Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
```

ipconfig /all

This command displays detailed information about your TCP/IP connection including router, gateway, DNS,DHCP and the type of Ethernet adaptor in your system.

```
C:\>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : DESKTOP-IPENDKP
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No
```

ipconfig /renew

This command will renew all your IP addresses that you are currently borrowing from DHCP server.

ipconfig /release

This command allows you to drop the IP lease from the DHCP server.

ipconfig /flushdns

This command is used to clear DNS cache.

c. **hostname:** This command displays the name of your computer.



d. **netdiag:**

This is a network testing utility that performs a variety of network diagnostic tests that allows you to diagnose problems in your network.

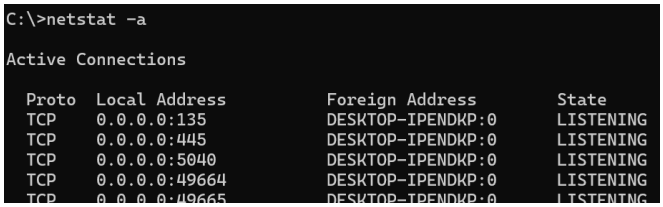
netdiag /test:TestName

netdiag /l

e. **netstat:**

The command stands for **network statistics**. It is used for displaying network connections for TCP, routing tables and various other network interface statistics. It is available on Linux, Solaris, Windows and BSD. You would typically use it to troubleshoot network issues, determine traffic details and measure performance.

Syntax: netstat -a



f. **nslookup:**

The command stands for **nameserver lookup**. It is used to query DNS domain names or IP address mappings. It is available on any operating system. A non-authoritative answer means that the response came from a cached entry and not the authoritative server for that domain.

Syntax: nslookup IP-Address



g. **pathping:** Pathping is a Windows based command used to trace the route to destination address, then launch a 25 second test of each router along the way, gathering statistics on the rate of data loss along each hop.

```
C:\>pathping -g www.google.com

Tracing route to www.google.com [142.250.76.196]
over a maximum of 30 hops:
  0  DESKTOP-IPENDKP [192.168.29.32]
  1     *        *        *
Computing statistics for 0 seconds...
            Source to Here   This Node/Link
Hop  RTT    Lost/Sent = Pct  Lost/Sent = Pct  Address
  0                                          DESKTOP-IPENDKP [192.168.29.32]

Trace complete.
```

h. **ping:** Ping stands for **Packet Internet Groper**. It is a TCP/IP service that runs on ICMP (Internet Control Message Protocol) and reports on the IP connectivity of network devices. Firewalls and ACLs can block ICMP and hence failure of ping might not actually mean the device is down.

```
C:\>ping amazon.com

Pinging amazon.com [52.94.236.248] with 32 bytes of data:
Reply from 52.94.236.248: bytes=32 time=223ms TTL=231
Reply from 52.94.236.248: bytes=32 time=319ms TTL=231
Reply from 52.94.236.248: bytes=32 time=219ms TTL=231
Reply from 52.94.236.248: bytes=32 time=305ms TTL=231

Ping statistics for 52.94.236.248:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 219ms, Maximum = 319ms, Average = 266ms

C:\>ping 52.94.236.248

Pinging 52.94.236.248 with 32 bytes of data:
Reply from 52.94.236.248: bytes=32 time=273ms TTL=231
Reply from 52.94.236.248: bytes=32 time=276ms TTL=231
Reply from 52.94.236.248: bytes=32 time=224ms TTL=231
Reply from 52.94.236.248: bytes=32 time=266ms TTL=231

Ping statistics for 52.94.236.248:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 224ms, Maximum = 276ms, Average = 259ms
```

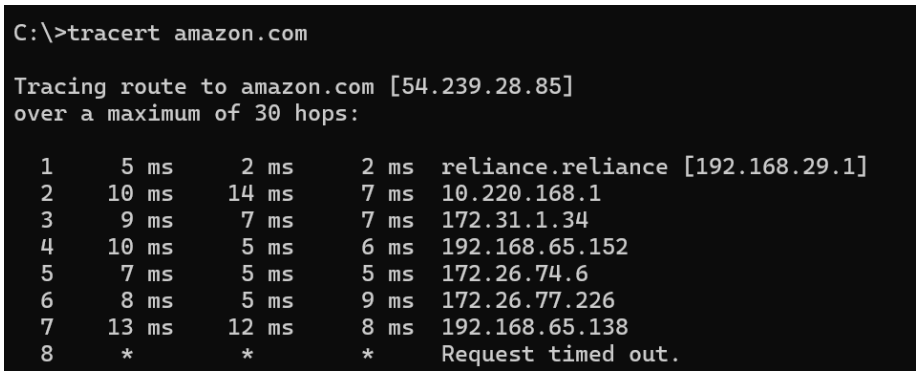
i. **route:** This command displays computer's routing table.

```
C:\>route PRINT 192*
===========================================================================
Interface List
 15...a8 59 5f c3 d6 9d ......Microsoft Wi-Fi Direct Virtual Adapter
 16...aa 59 5f c3 d6 9c ......Microsoft Wi-Fi Direct Virtual Adapter #2
  6...a8 59 5f c3 d6 9c ......Intel(R) Wi-Fi 6E AX211 160MHz
 18...a8 59 5f c3 d6 a0 ......Bluetooth Device (Personal Area Network)
  1...........................Software Loopback Interface 1
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
     192.168.29.0    255.255.255.0         On-link     192.168.29.32    306
    192.168.29.32  255.255.255.255         On-link     192.168.29.32    306
   192.168.29.255  255.255.255.255         On-link     192.168.29.32    306
===========================================================================
Persistent Routes:
  None

IPv6 Route Table
===========================================================================
Active Routes:
  None
Persistent Routes:
  None
```

j. **tracert:** This command displays a list of all routers that a packet has to go through to get from a computer where tracert is run to any other computer on the internet.



```
C:\>tracert amazon.com

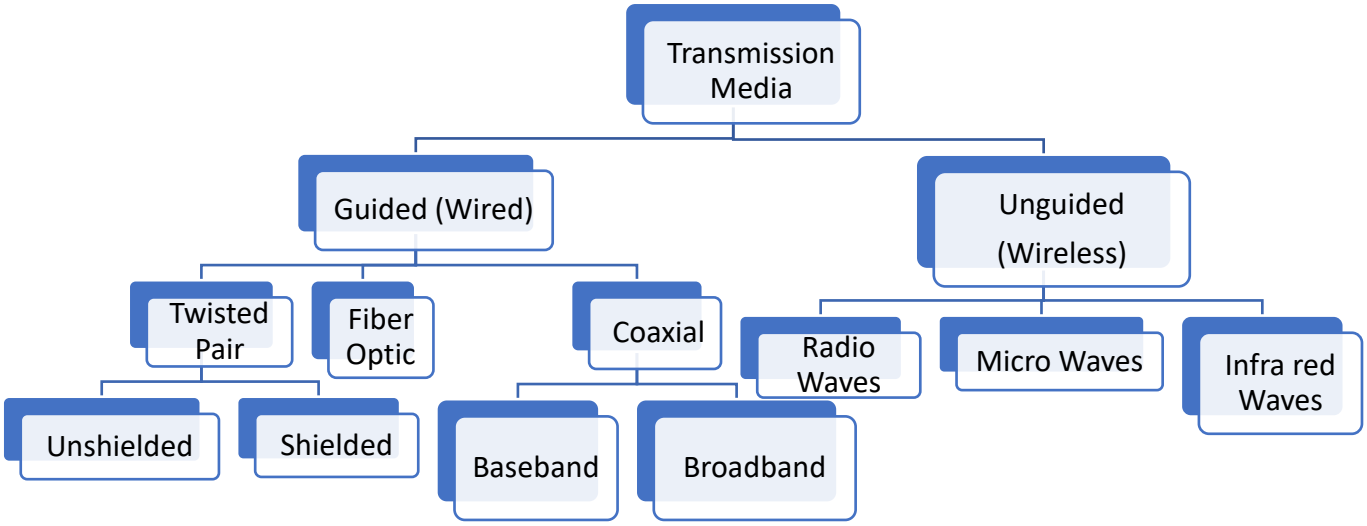
Tracing route to amazon.com [54.239.28.85]
over a maximum of 30 hops:

  1     5 ms     2 ms     2 ms  reliance.reliance [192.168.29.1]
  2    10 ms    14 ms     7 ms  10.220.168.1
  3     9 ms     7 ms     7 ms  172.31.1.34
  4    10 ms     5 ms     6 ms  192.168.65.152
  5     7 ms     5 ms     5 ms  172.26.74.6
  6     8 ms     5 ms     9 ms  172.26.77.226
  7    13 ms    12 ms     8 ms  192.168.65.138
  8     *        *        *     Request timed out.
```

2. Study of different types of Network Cables.

Transmission Medium:

A Transmission medium can be broadly defined as anything that can carry information from a source to a destination.

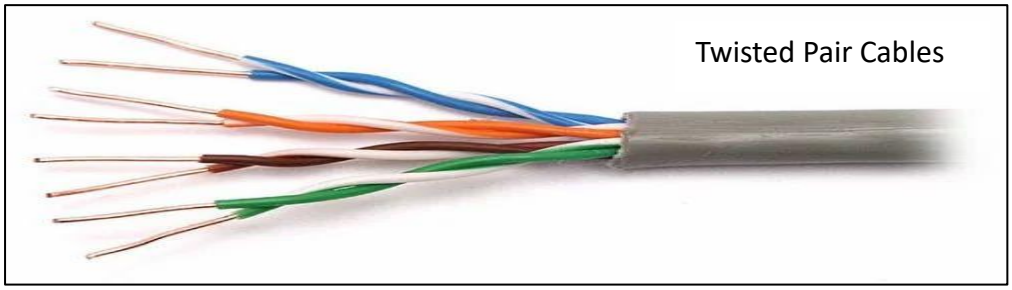


Guided Transmission Media:

This kind of transmission media is also known as wired otherwise bounded media. In this type, the signals can be transmitted directly & restricted in a thin path through physical links. The types of Bounded /Guided transmission are discussed below.

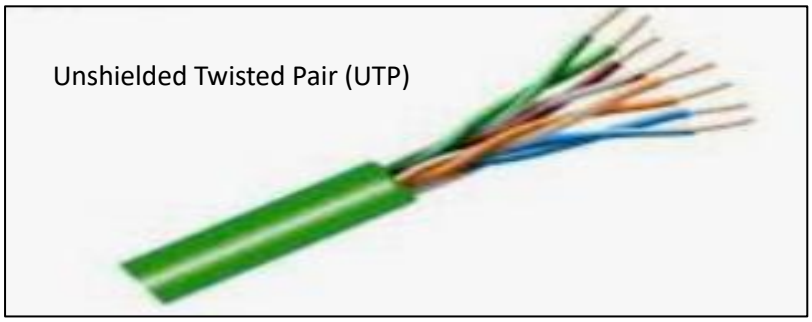
a. Twisted Pair:

This type consists of two conductors (normally Copper), each with its own plastic insulation, twisted together. One of the wires is used to carry signals to the receiver, while other is used as a ground reference only. By twisting the pairs, a balance is maintained.



i) UnShielded Twisted Pair (UTP):

This type is widely used in telecommunication. As there is no metal shield, its quality is average and hence less expensive but most commonly used.



UTP Categories - Copper Cable				
UTP Category	Data Rate	Max. Length	Cable Type	Application
CAT1	Up to 1Mbps	-	Twisted Pair	Old Telephone Cable
CAT2	Up to 4Mbps	-	Twisted Pair	Token Ring Networks
CAT3	Up to 10Mbps	100m	Twisted Pair	Token Rink & 10BASE-T Ethernet
CAT4	Up to 16Mbps	100m	Twisted Pair	Token Ring Networks
CAT5	Up to 100Mbps	100m	Twisted Pair	Ethernet, FastEthernet, Token Ring
CAT5e	Up to 1 Gbps	100m	Twisted Pair	Ethernet, FastEthernet, Gigabit Ethernet
CAT6	Up to 10Gbps	100m	Twisted Pair	GigabitEthernet, 10G Ethernet (55 meters)
CAT6a	Up to 10Gbps	100m	Twisted Pair	GigabitEthernet, 10G Ethernet (55 meters)
CAT7	Up to 10Gbps	100m	Twisted Pair	GigabitEthernet, 10G Ethernet (100 meters)

Advantages:

- Inexpensive and readily available.
- Flexible and light weight.
- Easy to setup and install.

Disadvantages:

- Susceptible to interference and noise.
- Low Bandwidth.
- It can be used only for shorted distances because of attenuation.

Applications:

- Used in telephone lines to provide Voice and data channels.
- Used in DSL lines.
- LANs such as 10Base-T and 100Base-T.

ii) Shielded Twisted Pair(STP):

Shielded Twisted Pair cables additionally have an overall conducting metallic shield covering four twisted pair wires. There could be a metallic shield covering individual twisted pairs also. These metallic shields block out electromagnetic interference to prevent unwanted noise from the communication circuit.



Shielded Twisted Pair

Advantages:

- Improved quality due to metal casing.
- Offers lower attenuation and better noise resistance as compared to UTP.
- Allows for longer cable runs.
- Offers higher data rates in noisy environments.

Disadvantages:

- Bulkier and expensive
- Difficult installation due to stiffness
- Needs proper grounding to avoid certain types of electromagnetic interference.

Applications:

- Industrial Control Systems
- High Speed data networks for servers and storage.
- Linking Surveillance systems.
- Connecting sensitive medical devices like imaging systems.

b. Coaxial Cable:

Coaxial cable has a central core conductor of solid wire which is enclosed in insulation, which in turn is encased in an outer conductor of metal foil. The outer metallic serves as a shield against noise and also serves as a second conductor. This is again enclosed in insulation and the whole cable is protected by plastic cover. To connect coaxial cable to devices, we need coaxial connectors such as BNC connector.

Advantages:

- Offers higher bandwidth enabling faster data transfer for applications like cable TV and internet.
- Superior Shielding that ensures clean and reliable signal.
- Blocks external electromagnetic interference and noise preventing signal degradation.
- Durable and resistant to damage from crushing, cutting and bending.
- Capable of carrying multiple channels simultaneously.

Disadvantages:

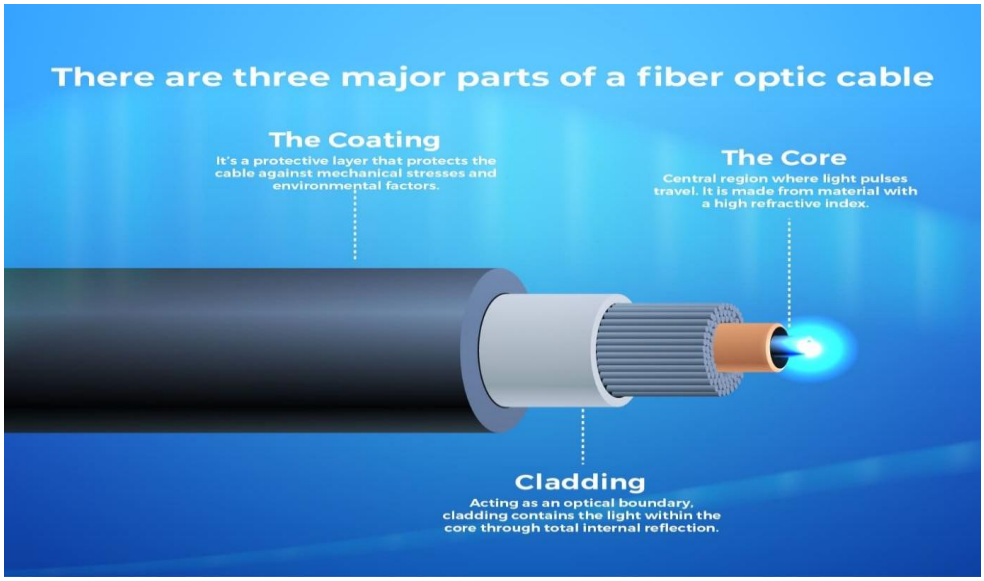
- Attenuation is much higher in Coax as compared to twisted pair.
- The signal weakens rapidly in coax and requires frequent use of repeaters.
- High installation and maintenance costs.
- Bulky and inflexible design.

Applications:

- Used earlier in Analog and Digital networks.
- Traditional Cable TV networks.
- Traditional Ethernet LANs.

c. Fiber Optic Cables:

These cables are made of glass or plastic and transmits signals in the form of light. It uses reflection to guide light through a channel. A glass or plastic core is surrounded by a cladding of less dense glass or plastic. The difference in the density of the two materials causes this reflection of beam of light.



Advantages:

- Higher Bandwidth
- Less signal attenuation. In fact, it is flatter than twisted pair or coaxial cable. We need only fewer repeaters.
- Immunity to electromagnetic interference.
- Resistance to corrosive materials
- Light weight
- Greater immunity to tapping.

Disadvantages:

- Installation and maintenance
- Unidirectional light propagation makes it complicated for bidirectional communication.
- High cost due to specialized equipment and skilled technicians required for the job.
- Delicate and fragile equipment makes them prone to damage.

Applications:

- Used in backbone networks because of its wide bandwidth and cost effectiveness.
- LAN such as 100Base-Fx network
- High speed internet, cable TV and telephone networks.
- Used in medical field for endoscopy and imaging.
- Security focused military, aerospace and industrial sensing applications.

Unguided Transmission Media:

Unguided medium transport electromagnetic waves without using a physical conductor. This is referred to as wireless communication. Different types of Unguided media are discussed below.

a. Radio waves:
   These are electromagnetic waves ranging in frequencies between 3KHz to 1GHz. They are omni directional, which means, they can travel in all directions. They are used for multicast communication such as radio, television and paging systems.
b. Micro waves:
   These are electromagnetic waves having frequencies between 1 to 300 GHz. They are unidirectional, meaning their propagation is line of sight. Repeaters are often needed for long distance communication. Very high frequency microwaves cannot penetrate walls. They are used in unicast communication such as Cellular telephones, Satellite networks and wireless LANs.

c. Infrared Waves:
   These are electromagnetic waves with frequencies 300GHz to 400THz. These are used for short range communication and cannot penetrate walls if frequency is high. They are used in TV remote, signals for communication between devices such as keyboards, mouse, PC and Printers.

3. Practically implement the cross wired cable and straight wired cable using Crimping tool.

Requirements:

Crimping tool, UTP cable, RJ-45 connector, Cable tester.

Understanding of tools used:

a. RJ-45:

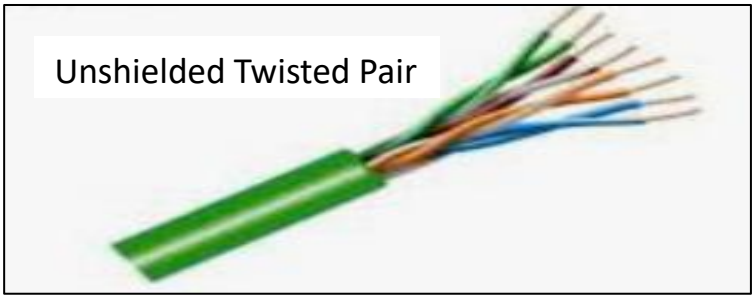
A registered jack (RJ) is a standardized physical network interface for connecting telecommunications or data equipment. The most common twisted pair connector is the 8position, 8contact (8P8C) modular plug and jack commonly referred to as an RJ-45 connector. The 8-pin RJ-45 connector is commonly used to connect computers onto Ethernet based local area network (LAN).



RJ-45

b. UTP Cable:

Unshielded Twisted Pair (UTP) copper cable has no metallic shield, because of which its diameter is small. The twist helps to improve its immunity to electrical noise and external electromagnetic interference.



c. Crimping tool:

RJ-45 Crimping tool is a 3-in-1 tool used to strip, cut and crimp cable, all in a single tool. It is a device used to conjoin two pieces of metal by deforming one or both of them to hold each other.



Crimping Tool

d. Cable Tester:

A Cable tester is an electronic device used to verify electrical connections in a signal cable or other wired assembly.



Procedure:

1. Outer Sheathe Stripping:

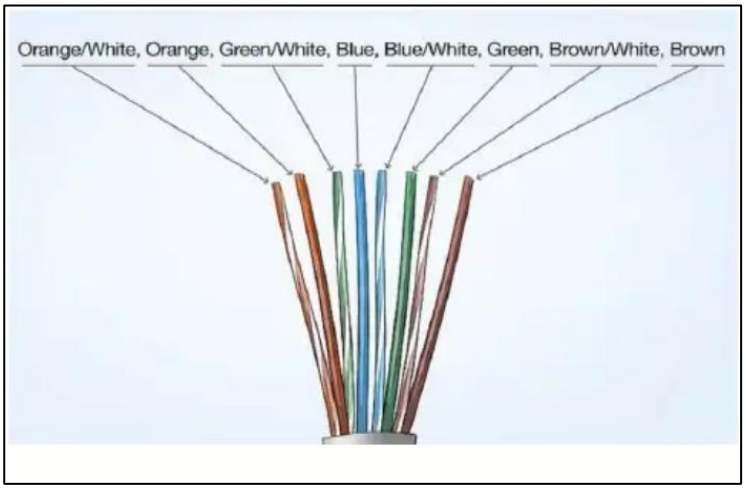
   Insert the cable 1'' from the end into the stripper portion of the crimping tool and squeeze it tight. While squeezed, rotate the crimp tool around the cable a full 360^0. Do not cut the twisted pair wires under the outer sheathe. This may result in a decrease in or no data transfer.



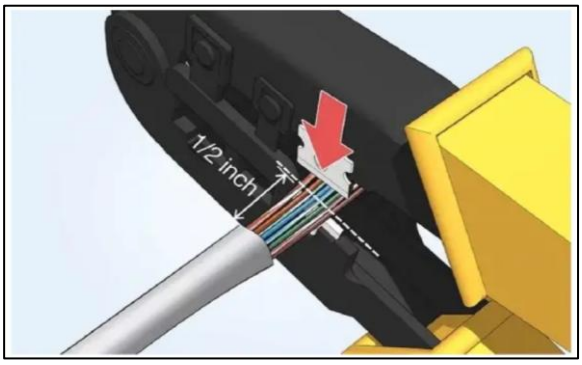
2. Wire untwisting and 568B scheme arrangement:

   Untwist the smaller wires and arrange them into proper wiring scheme (568B – the recommended one) for the RJ-45 connector. The scheme is as follows:

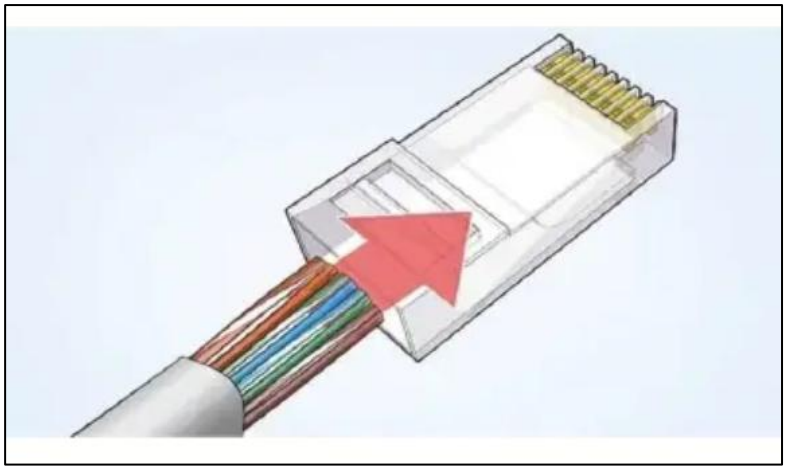
Pin 1 – Orange/White
Pin 2 – Orange
Pin 3 – Green/White
Pin 4 – Blue
Pin 5 – Blue/White
Pin 6 – Green
Pin 7 – Brown/White
Pin 8 – Brown



3. Wire preparation for RJ-45 Connector:
   Bring the wires tighter together and cut them down, in an even line, to ½'' from the cut of the sheathing using the 3-in-1 crimping tool.

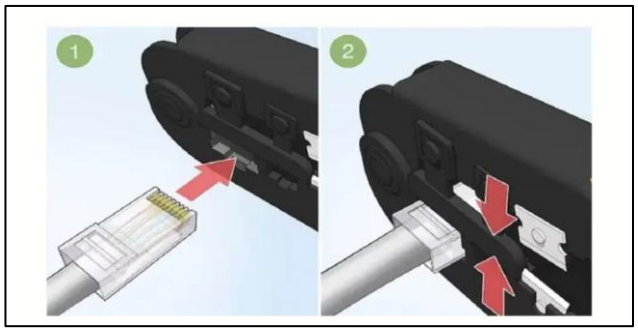


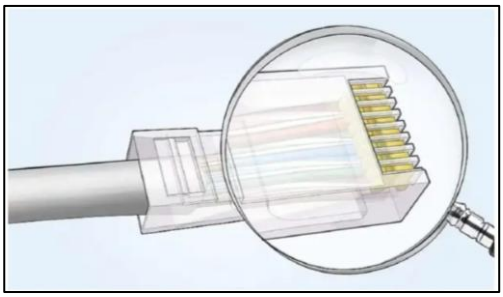
4. Inserting wire into the RJ-45 Connector:
   With the RJ-45 connector facing up (securing clip on the underside), insert the wires into the connector. Each wire will fit into each of the eight grooves of the connector. The wires should be easily visible looking into the end of the connector. Improper insertion may result in improper crimping of the pins to the wire.



5. Testing proper Crimp:

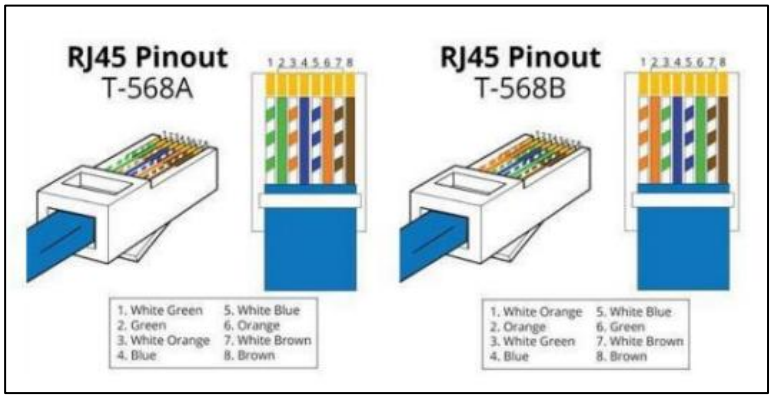
   Finally, the RJ-45 connector needs to be crimped onto the wire. When this happens, the 8 pins are pushed down into the wires below.
   Insert the connector into the crimping portion of the tool until the connector cannot go any further. Squeeze the crimping tool very tightly and release.
   When the crimping is complete, remove the wire from the tool and check the pins to make sure that they are all down.





Procedure for Straight Cable:

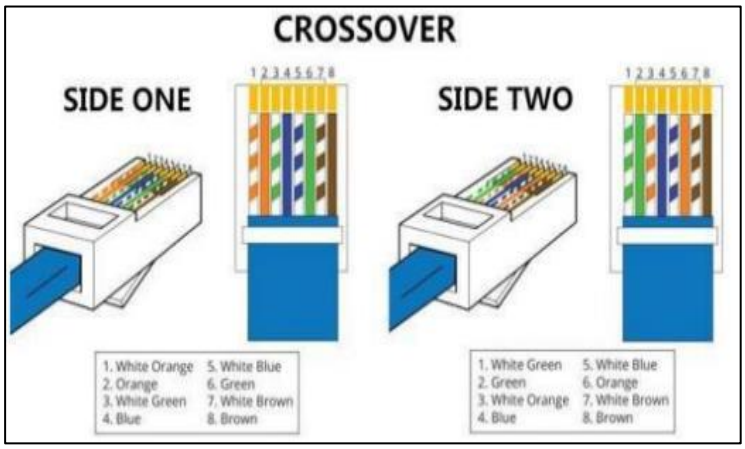
Straight-through cable is a type of CAT5 with RJ—45 connectors at each end and each has the same pin out. This type of twisted pair cable is used in LAN to connect a computer to a network hub such as a router. This cable uses one wiring standard – both ends use either T568A or T568B standard.



Procedure for Crossover Cable:

A Crossover cable is a type of CAT5 where one end is T568A configuration and the other is T568B configuration. Here, Pin 1 is crossed with Pin 3 and pin 2 is crossed with Pin 6. The internal wiring of the crossover cables reverses the transmission and receive signals. It is used to connect two or more computing devices (two devices of the same type).

Ex: Connecting two computers or two switches together.



4. Study of network IP address configuration (Classification of address, Static and dynamic).

The IP Address stands for Internet Protocol address. It is a numerical label assigned to the devices connected to a computer network that uses IP for communication. Thus, an IP address acts as an identifier for a specific machine on a particular network. It also helps to develop a virtual connection between a source and a destination.

Types of IP Address:

There are mainly 4 types of IP addresses:

- Public
- Private
- Static
- Dynamic

Public IP Addresses:

A public IP address is an address where one primary address is associated with the whole network. Each of the connected device has the same IP address. Such an IP address I provided by an Internet Service Provider (ISP).

Private IP Addresses:

A private IP address is a unique IP number assigned to every device that connects to internet network, which includes devices such as computers, tablets, smartphones, etc.

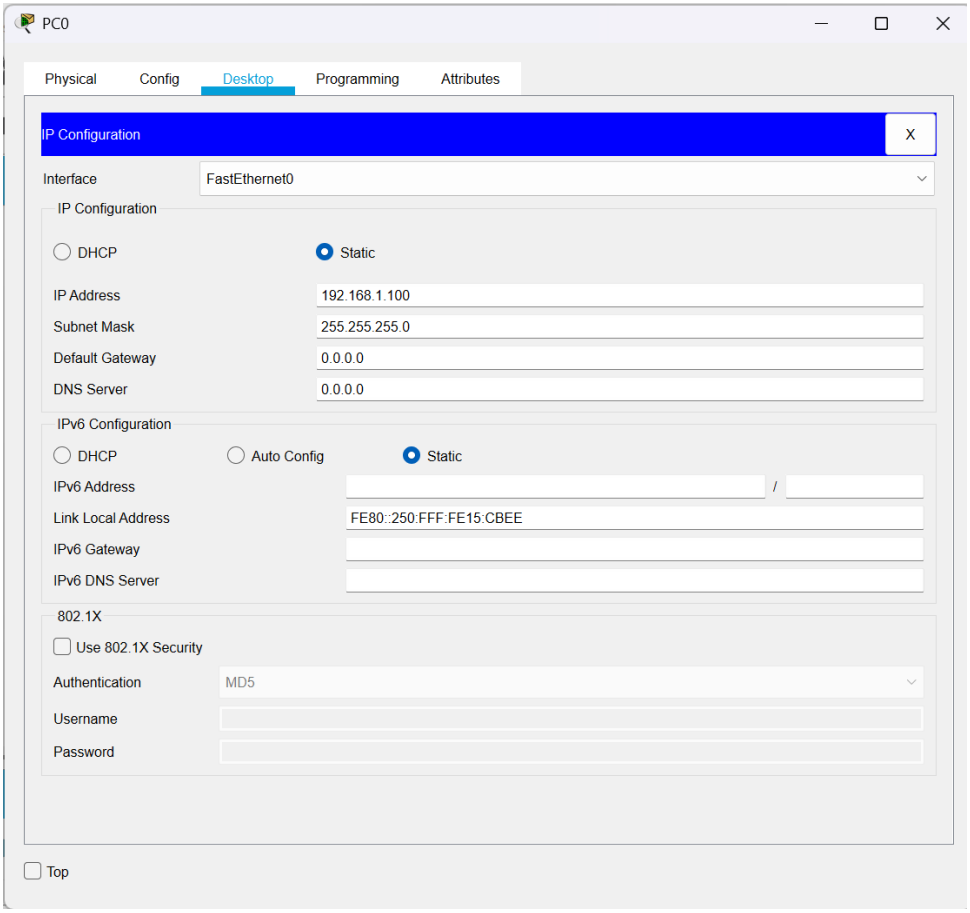
Static and Dynamic IP Addresses:

The main differences between Static and Dynamic IP addresses are listed below:

SlNo.	Features	Static IP Address	Dynamic IP Address
1.	Definition	It is a permanent numeric address that is manually issued to a network device.	It is a temporary IP address allocated to a system when it connects to a network.
2.	Provider	It is provided by the Internet Service Provider.	It is provided by DHCP(Dynamic Host Control Protocol).
3.	Changes	It doesn't change with time.	It may be changed at any time.
4.	Device tracking	Devices may be easily traced.	Devices may be difficult to trace.
5.	Cost	It is expensive to utilize and maintain.	It is less expensive to utilize and maintain.
6.	Security	It is less secure than dynamic IP address.	It offers high security.
7.	Designation	It is complex to assign and reassign.	It is much easy to assign and reassign.
8.	Stability	It is highly stable.	It is less stable.
9.	Usage	These are appropriate for dedicated services like FTP, mail and VPN servers.	These are appropriate for a large network that needs an internet connection for all devices.

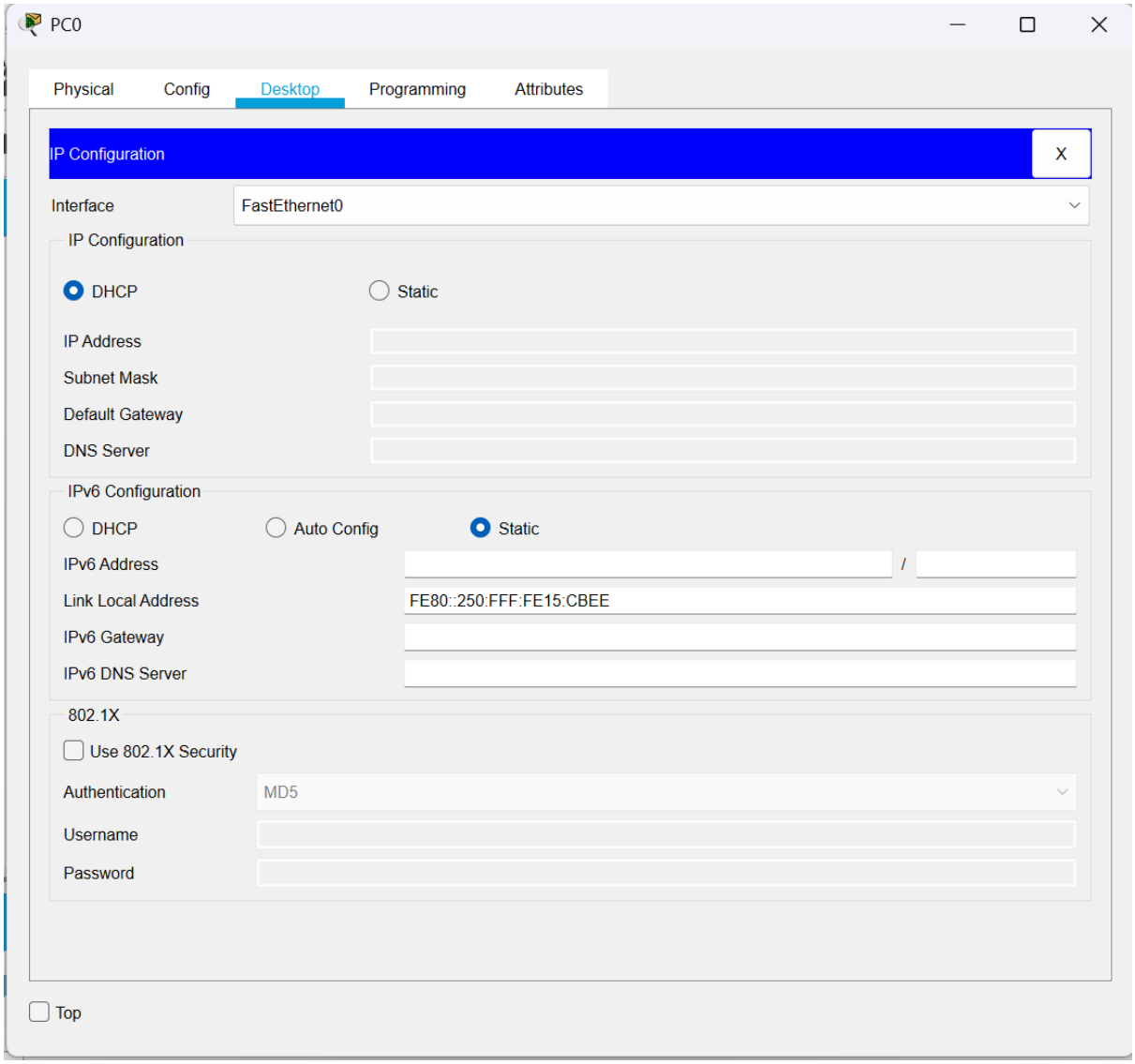
Configuration of Static IP Address:

1. Drag and Drop PC icon under end devices, onto the workplace, in the Cisco packet tracer.
2. Double click on the PC icon to edit its properties. Under Desktop->IP Configuration, set the static IP Address manually as shown below:



Configuration of Dynamic IP Address:

1. Drag and Drop PC icon under end devices, onto the workplace, in the Cisco packet tracer.
2. Double click on the PC icon to edit its properties. Under Desktop->IP Configuration, select DHCP option to set Dynamic IP address as shown below:



5. Study the Network IP Address Configuration (Classification as IPv4 and IPv6, Subnet, Supernet).

The Internet Protocol Version 4 (IPv4) is a protocol used on packet switched linked layer networks (ex: Ethernet). It provides an addressing capability of approximately 4.3 billion addresses. The Internet Protocol version 6 (IPv6) is more advanced and has better features compared to IPv4.

Feature	IPv4	IPv6
Address Length	32-bit address	128-bit address
Fields	Numeric address that consists of 4 fields separated by dot (**.**).	Alphanumeric address that consists of 8 fields, separated by colon(**:**)
Classes	5 different classes of IP address that include class A, class B, class C, class D and class E.	Does not contain any class for IP address.
Number of IP addresses	Limited	Supports a large number of IP addresses.
VLSM (Virtual Length Subnet Mask – conversion of IP address into subnet of different sizes)	Supports VLSM.	Does not support VLSM.
Address Configuration	Supports manual and DHCP configuration.	Supports manual, DHCP, auto configuration and renumbering.
Address Space	It generates 4 billion unique addresses.	It generates 340 undecillion unique addresses.
End to End connection integrity.	Unachievable	Achievable
Security Features	IPv4 is not developed keeping security feature in mind. It is up to the application to handle security.	IPSEC is developed for Security purposes.
Address representation	Address is represented in decimal.	Address is represented in hexadecimal.
Fragmentation	Done by senders and forwarding routers.	Done by only Senders.
Packet flow identification	No mechanism provided	It uses flow label field in the header for packet flow identification.
Checksum field	Available	Not available
Transmission Scheme	Broadcasting	Multicasting, which provides efficient network operations.
Encryption and Authentication	Not provided	Provided
Number of Octets	4	16 (8 fields, each with 2 octets).

Subnetting is a technique of partitioning an individual physical network into several small sized logical sub-networks. These subnetworks are known as **Subnets**. An IP Address is made up of combination of network segment and a host segment. Subnetting converts host bits into network bits. This strategy was developed for slowing down the depletion of IP addresses.

Subnetting permits the administrator to partition a single class A, class B and class C network into smaller parts. VLSM is a technique which partitions IP address space into subnets of different sizes and prevent memory wastage. When number of hosts is same in the subnets, then it is known as **FLSM** (**Fixed length Subnet Mask**).

Subnetted Address: 192.168.1.0/24

In Binary: 11000000.10101000.00000001.00000000

1st Subnet – 192 . 168 . 0000 0001 . 00 000000 = 192.168.1.0/26

2nd Subnet – 192 . 168 . 0000 0001 . 01 000000 = 192.168.1.64/26

3rd Subnet – 192 . 168 . 0000 0001 . 10 000000 = 192.168.1.128/26

4th Subnet – 192 . 168 . 0000 0001 . 11 000000 = 192.168.1.192/26

5th Subnet – 192 . 168 . 0000 0010 . 00 000000 = 192.168.2.0/26

Changing bits

Supernetting is the inverse process of subnetting in which several networks are merged into a single network. This process is also known as **router summarization and aggregation**. It results in the creation of most host addresses at the expense of network addresses where the network bits are converted into host bits.

Supernetting is performed by the internet service provider to achieve most efficient IP address allocation. **CIDR** (**Classless Inter Domain Routing**) is the scheme used to route network traffic across the internet.

Supernetted Address: 192.168.1.0/24

In Binary: 11000000.10101000.00000001.00000000

192.168.1.0/24 = 192 . 168 . 00000 001 . 00000000

192.168.2.0/24 = 192 . 168 . 00000 010 . 00000000

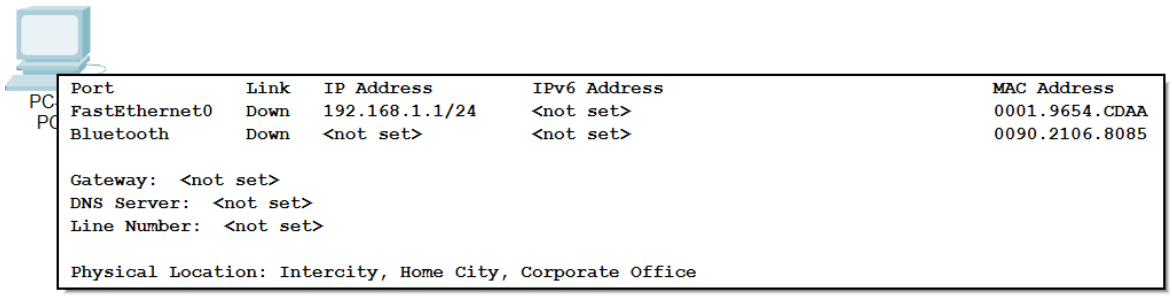
192.168.3.0/24 = 192 . 168 . 00000 011 . 00000000

192.168.4.0/24 = 192 . 168 . 00000 100 . 00000000

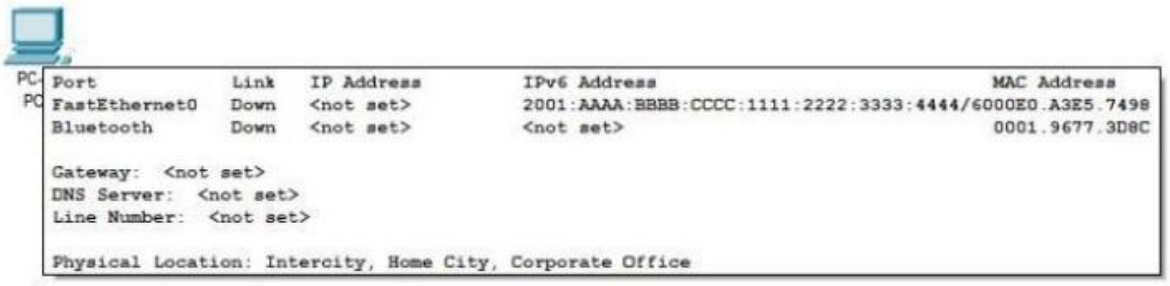
192.168.5.0/24 = 192 . 168 . 00000 101 . 00000000

No. of common bits = 21 No. of uncommon bits = 11

IPv4:



IPv6:



6: Study of Network Devices (Switch, Router and Bridge)

Software/Hardware Requirements: No software/Hardware requirements needed.

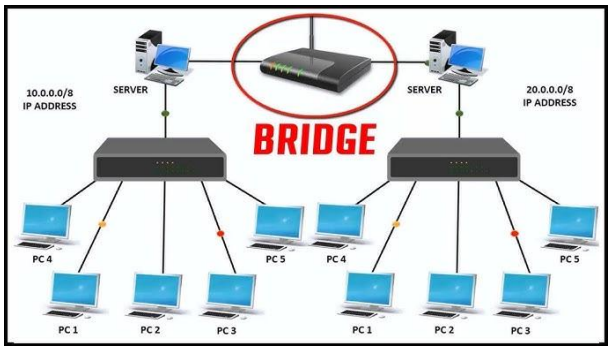
1. **Switch**: A network switch or switching hub is a computer networking device that connects network segments. It processes and routes data at the data link layer (layer 2) of the OSI model. Switches that additionally process data at the network layer (layer 3 and above) are referred to as Layer-3 switches or Multilayer Switches.
   Multiple data cables are plugged into a switch to enable communication between different networked devices. A switch manages flow of data across a network by transmitting a received network packet only onto one or more devices for which the packet is intended.





Working of a Switch: When a switch is initialized, the switch does not contain any information about any host or its address. When a host sends a frame, its MAC address along with port number will be stored in the address table of the switch. As and when hosts start sending data to the switch, the switch completes its address table. When a data packet arrives at a switch, it reads the destination MAC address, checks its internal MAC address table and sends the packet to only that port.

2. **Bridge**: A network bridge connects multiple network segments at the data link layer (Layer 2) of the OSI model. In Ethernet, the term bridge formally means a device that behaves according to the IEEE 802.1D standard. A switch is a bridge with numerous ports. A bridge analyzes incoming data packets to determine if it can send the given packet to another segment of the network. It is a passive device as there is no interaction between the bridge and the paths of bridging. It is used to lower the load of traffic on the LAN and other network.



Working of a Bridge: When the packets of data are transferred along a network, without locating the network addresses, the process is called **bridging**. It is helpful in locating the addresses of unknown addresses to which the bridge is viable to send data. When the process of bridging occurs, the bridge makes a bridging table alongside, where it stores the MAC addresses of various terminals. This table helps the bridges to send the data packet to the exact location next time.

There are two types of bridging modes:

Transparent Bridging: when a specific address does not match the contents of the bridging table, the data packet is forwarded to every attached terminal in the LAN except the terminal from which it received the packet.

Source Routing Bridging: When the source computer presents pathway information within the packet, then it is called Source routing bridging. This type is commonly used in Token Ring networks.

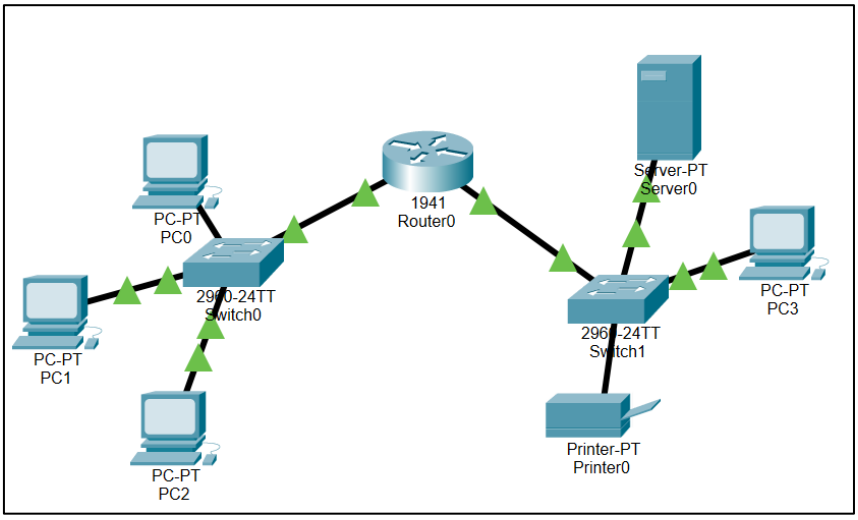
3. **Router:**

A Router is an electronic device that interconnects two or more computer networks, and selectively interchanges packets of data between them. It operates at layer 3 or the network layer of the OSI model. It is responsible for receiving, analyzing and forwarding data packets among the connected computer networks. When a data packet arrives, the router inspects the destination address, consults its routing tables to decide the optimal route and then transfers the packet along the route.



Working of a Router:

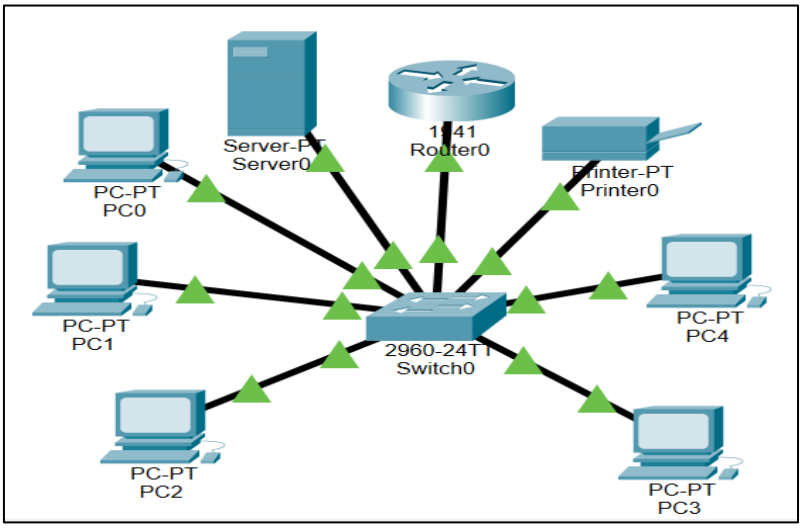
A router examines a packet header's destination IP address and compares it against a routing table to determine the packet's next best hop. Routing tables amount to an algorithmic set of rules that calculate the best way to transmit traffic toward any given IP address. A Routing table often specifies a default route, which the router uses whenever it fails to find a better forwarding option for a given packet. Routing tables can be static or dynamic. Dynamic routers automatically update their routing tables based on network activity, exchanging information with other devices via routing protocols.



7. Configure and connect the computer in LAN.

Follow the below mentioned procedure in Cisco Packet Tracer to connect various network devices in LAN.

a. Select 1 router, 1 switch, 5 PCs, 1 server and 1 printer through drag and drop.
b. Configure IP address for Router as 192.168.1.100 for the respective Ethernet connection. Double click on the router, Select the port status check box on. Connect the router to the switch using Straight through cable.
c. Configure Server IP address as 192.168.1.101 and connect to switch by a straight through cable. Click on the Server, select Desktop, IP Configuration.
d. Connect all 5 PCs to switch using Straight through cable and configure their IP addresses as 192.168.1.1 to 192.168.1.5.
e. Give gateway IP as 192.168.1.100 and DNS IP as 192.168.1.101. Click on PC, Select Desktop and click on IP Configuration.
f. Configure Printer IP 192.168.1.102 and connect to switch using straight through cable.



g. Ping all PCs, Server and Printer from any PC by issuing command, ping IP address.

```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.5

Pinging 192.168.1.5 with 32 bytes of data:

Reply from 192.168.1.5: bytes=32 time=1ms TTL=128
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.1.101

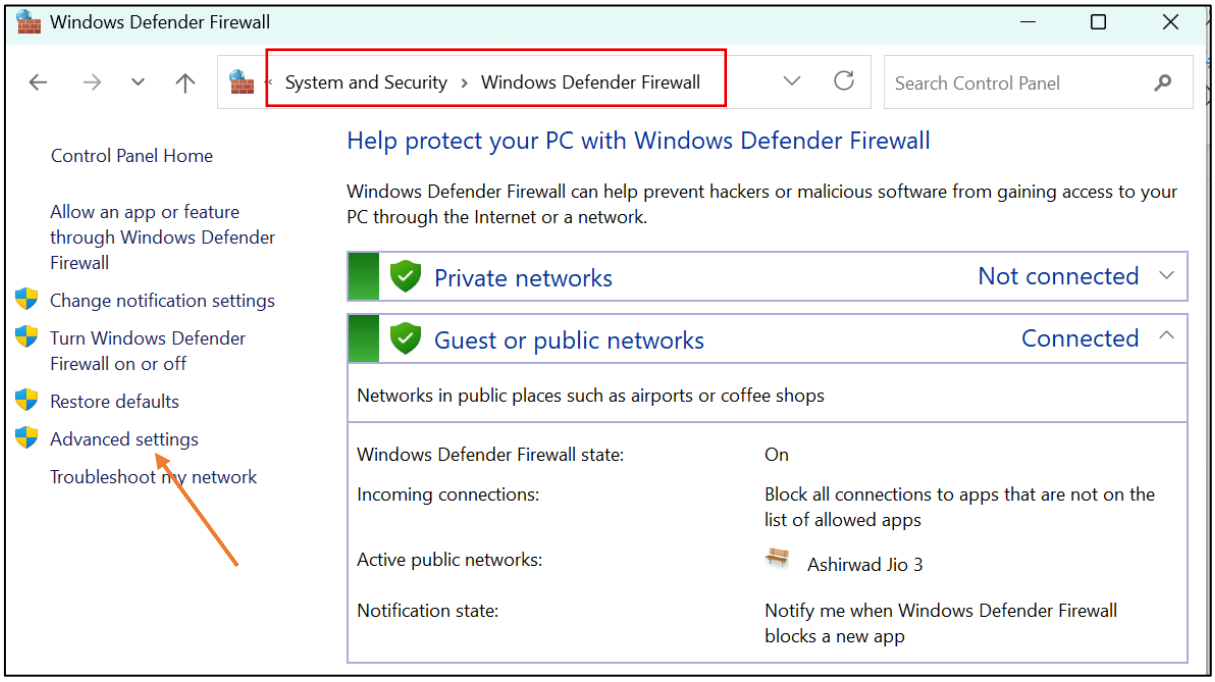
Pinging 192.168.1.101 with 32 bytes of data:

Reply from 192.168.1.101: bytes=32 time<1ms TTL=128
Reply from 192.168.1.101: bytes=32 time<1ms TTL=128
Reply from 192.168.1.101: bytes=32 time<1ms TTL=128
Reply from 192.168.1.101: bytes=32 time<1ms TTL=128
```

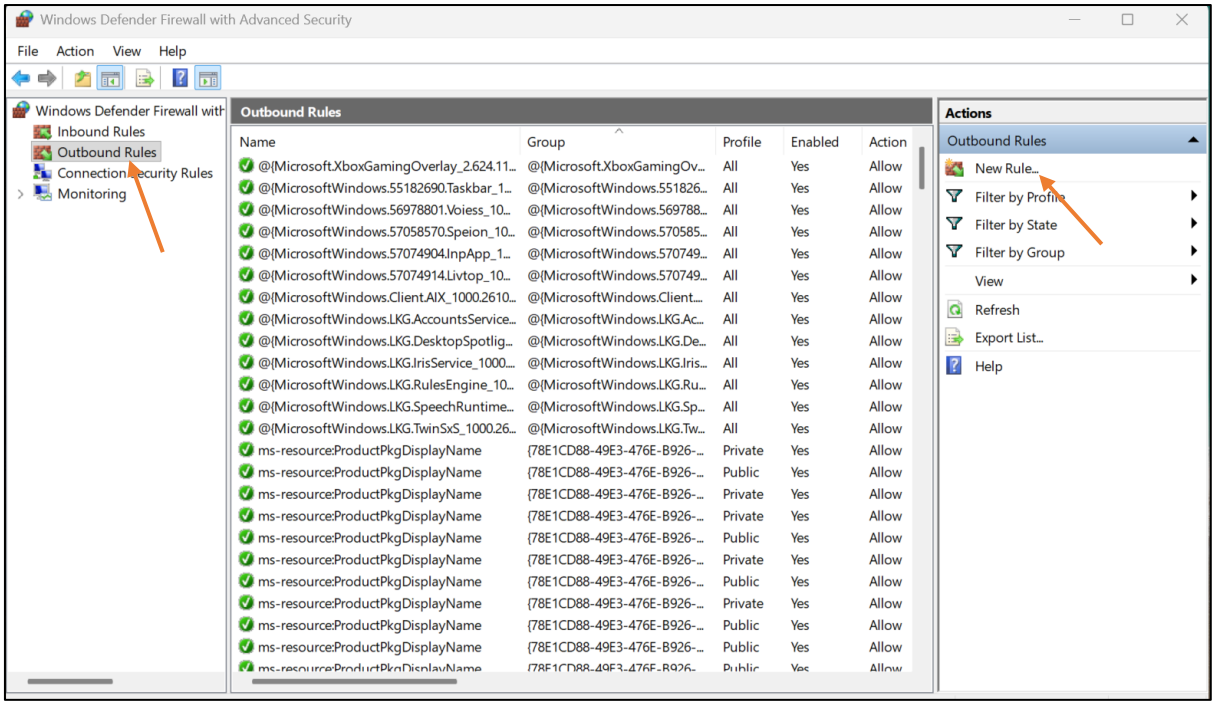
8. Block the website using “Windows Defender Firewall” in Windows 10.

Follow the below mentioned procedure in the control panel of your Windows 10 PC to connect complete this lab program.

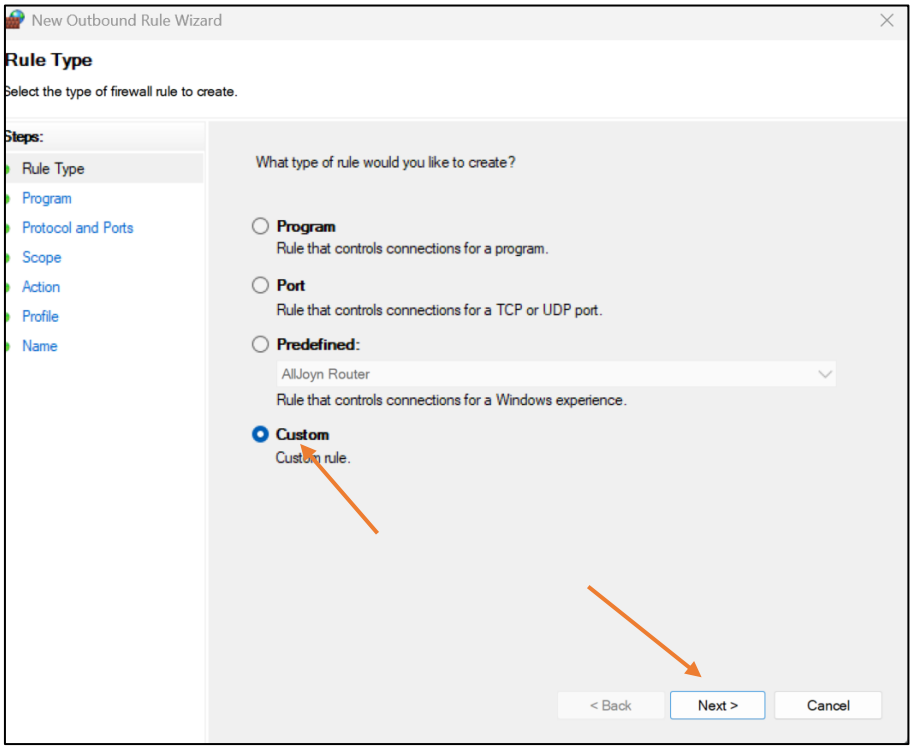
a. Launch the Control Panel in your Computer.
b. Select the “Windows Defender Firewall” (in some computers, under System and Security) followed by Advanced settings on the left side pane.



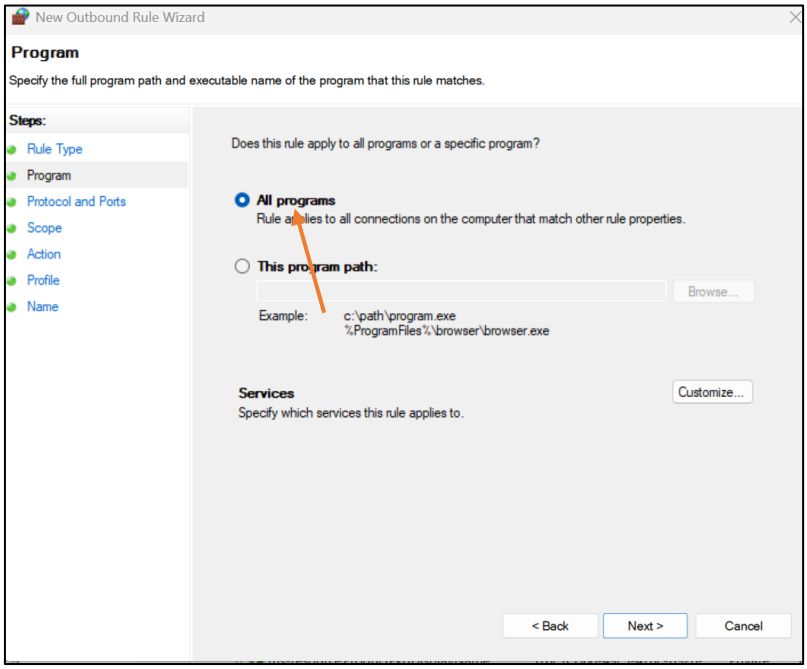
c. Right click on “Outbound Rules” from the Menu on the left and select “New Rule”.



d. When a new Window pops up, select “Custom” option followed by “Next”.



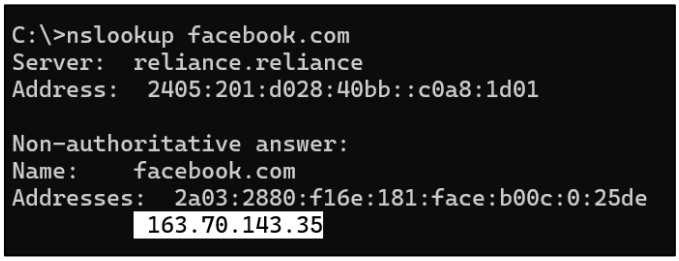
e. In the next window, select “All Programs” and click on “Next”.



f. Select the “These IP addresses” option under “Which remote IP addresses does this rule apply to?” and click Next.

g. Open Command Prompt as Administrator by entering CMD in the search box.

h. Enter “nslookup facebook.com” and press Enter.



i. Come back to the Windows Firewall window and click on Add and paste the IP address you want to block. Then select Next.

j. Choose “Block the connection” option and click Next.

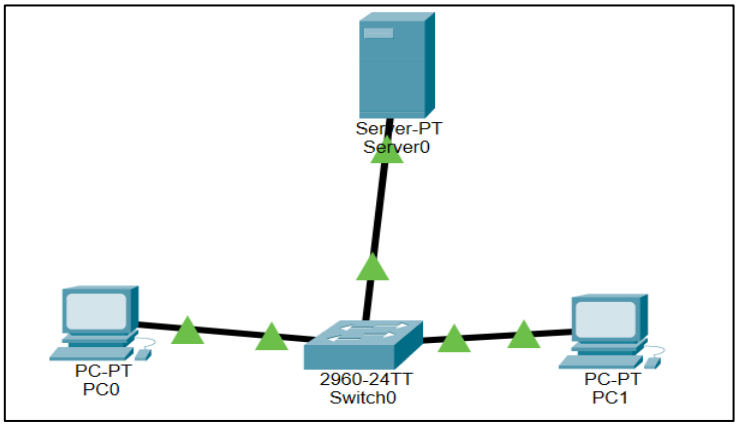
k. Choose Domain, Private and Public options for the rule to apply. Select Next.

l. Add a name or description for this rule and click “Finish” to complete the action.

m. Check for the Blocked Website.

9. Share the folder in a system and access the files of that folder from other system using IP address.

a. Create LAN server configuration as shown below (2 PCs, 1 Server and 1 switch):



b. Configure Server with IP address 192.168.1.100 and the two PCs with 192.168.1.1 and 192.168.1.2 respectively. For both PCs, and server, give DNS IP address as 192.168.1.101.

c. Click on Server -> Services -> FTP. Give Username and Password (username – admin, password – admin). Give the permission for admin (Write, Read, Delete, Rename, List). Click on Add.

d. On PC-0, create a text file, sampleText.txt – Click on PC-0 -> Desktop ->Text Editor -> File -> New.

e. Type some text and save the file as sampleText.txt.

f. In the same PC, click on command prompt. Check the newly created by entering,
C:\>dir
C:\>ftp 192.168.1.100 (server's IP address).
Give username and password
ftp>put sampleText.txt (file gets transferred from PC-0 to Server).
ftp>dir (Check if the file is being transferred to server).
ftp>quit

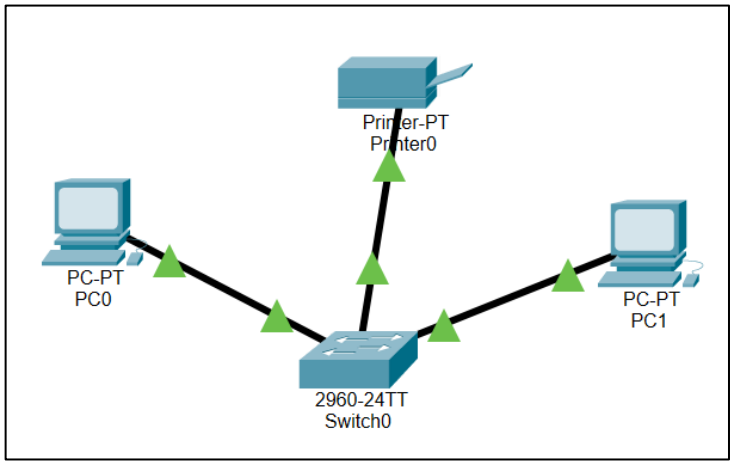
g. Select PC -1.
Click on Desktop -> Command Prompt
C:\>dir (check if file by name sampleText.txt is already present)
C:\>ftp 192.168.1.100 (server's IP address).
Give username and password
ftp>get sampleText.txt (file gets transferred from server to PC-1).
ftp>quit
C:\>dir
Verify to see the new file copied from Server through FTP.

10. Share the Printer in a network, and take a print from other PC.

a. Select 2 PCs, 1 Printer and 1 Switch through drag and drop.

b. Configure IP address to Printer as 192.168.1.3. Click on the Printer, Click on Config, Select FastEthernet0, Give IPv4 address and connect to switch using straight through cable.

c. Configure the IP addresses for the two PCs as 192.168.1.1 and 192.168.1.2 respectively. Connect the two PCs to the switch through straight through cable.



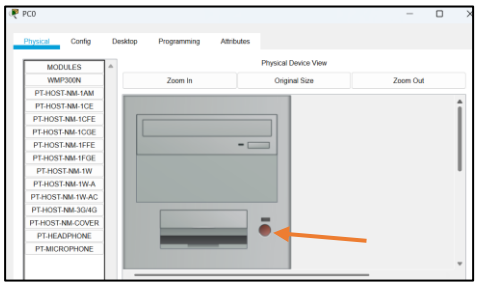
d. Ping Printer from any PC issuing ping command. Ex: ping 192.168.1.3

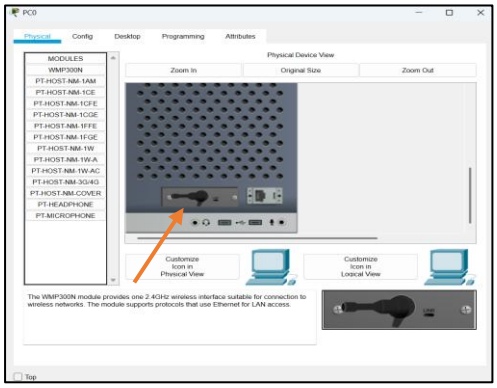
11. Configuration of Wi-fi Hotspot and connect other devices (Mobile/Laptop).

a. Select 2 PCs, 1 Server, 1 Laptop, 1 Printer, 1 smart phone, 1 home router by Drag and Drop to the workspace.

b. Configure IP address of the router.
   - Click on the Router -> LAN (give IP address to the router as 192.168.1.100).
   - Click on Wireless 2.4G
   - Click on WPA-PSK
   - Give the PSK pass phrase as 12345678
   - Change the SSID name from default to SSMRVBCA.

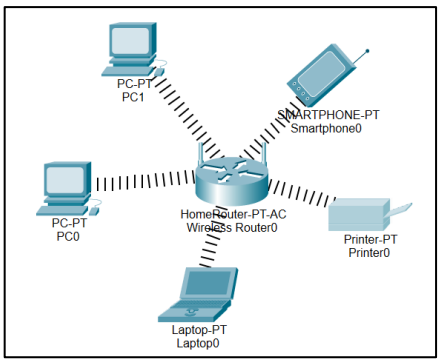
c. Make Wired PC to Wireless PC.
   - Double click on PC and select Physical.
   - Click on WMP300N (adaptor for wireless).
   - Switch off the PC.
   - Change the network LAN port to wireless LAN port through Drag and drop.
   - Switch on the PC.





d. Configure Wireless PC.
   - Click on Desktop -> DHCP -> config -> wireless0
   - Click on WPA-PSK and give PSK pass phrase as 12345678
   - Change the SSID name from default to SSMRVBCA.
   - PC will get connected to the router.

e. Repeat the above step to make Laptop, Printer, smartphone and server wireless.



f. Ping printer from any of the devices by issuing ping 192.168.1.105

12. Configuration of Switches

1. Drag and Drop a switch onto your workspace on Cisco Packet Tracer.

2. Configure Password for logging into switch:
   a) Click on the switch on go to CLI (Command line interface).
   b) Switch> EN (or ENABLE)
   c) Switch# config t (configuration terminal mode)
   d) Switch(config)#line con 0
   e) Switch(config-line)#password 123456
   f) Switch(config-line)#login
   g) Switch(config-line)#exit
   h) Switch(config)#exit
   i) Switch#exit

3. Configure Password for Configuration switch:
   a) Switch> EN (or ENABLE)
   b) Switch# config t (configuration terminal mode)
   c) Switch(config)# enable secret 12345678
   d) Switch(config)#exit
   e) Switch#exit

To check for Configuration password, first authenticate through console password.

User Access Verification

Password: (login password)
Switch>EN
Password: (Configuration password or Secret password)
Switch#

4. Configure switch hostname as **SSMRVBCA**
   a) Switch#configure t
   b) Switch(config)#hostname SSMRVBCA
   c) SSMRVBCA(config)#

5. Configure the message of the day as

"Welcome to SSMRV CISCO Switch Configuration"

SSMRVBCA(config)#banner motd #

"Welcome to SSMRV CISCO Switch Configuration"

----------#

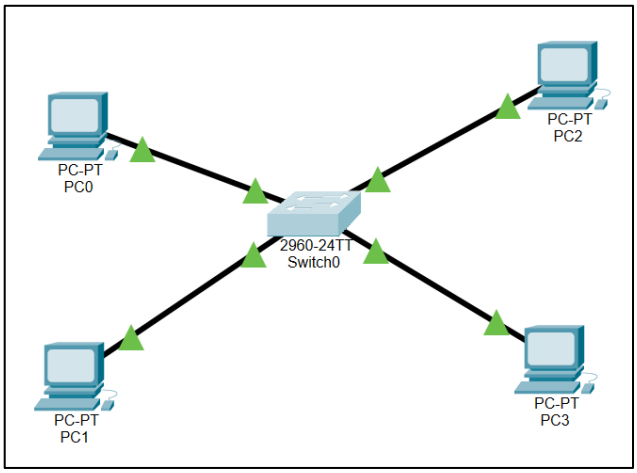
SSMRVBCA(config)#exit (To check the banner message, exit from configuration).

SSMRVBCA#exit

13. Configuration of VLan using Packet Tracer.

VLans help you segment your network for easier administration and added security. You will need a Layer 3 device (router) to ping between vlans.

Network Topology:



Step 1:
Drag and drop 4 PCs and a switch to your workplace on Packet Tracer using straight through cables.

Step 2:
Allocate IP addresses to the hosts from within the subnets they are assigned to:

VLAN 10 – 10.0.0.0 (10.0.0.1 and 10.0.0.2)
VLAN 20 – 20.0.0.0 (20.0.0.1 and 20.0.0.2)

Step 3:
Configure interfaces FastEthernet0/1 and FastEthernet0/2 into VLAN 10 and FastEthernet0/3 and FastEthernet0/4 into VLAN 20. Use switchport mode access command to force the ports to be access only and not trunk.

```
Switch>
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW1

SW1(config)#vlan 10
SW1(config-vlan)#name BCA
SW1(config-vlan)#exit
SW1(config)#vlan 20
SW1(config-vlan)#name BBA
SW1(config-vlan)#exit
SW1(config)#exit
SW1#
```

```
SW1(config)#vlan 10
SW1(config-vlan)#int fastEthernet0/1
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan 10
SW1(config-if)#int fastEthernet0/2
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan 10
SW1(config-if)#int fastEthernet0/3
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan 20
% Access VLAN does not exist. Creating vlan 20
SW1(config-if)#int fastEthernet0/4
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan 20
SW1(config-if)#exit
SW1(config)#exit
SW1#
%SYS-5-CONFIG_I: Configured from console by console
```

Step 4:

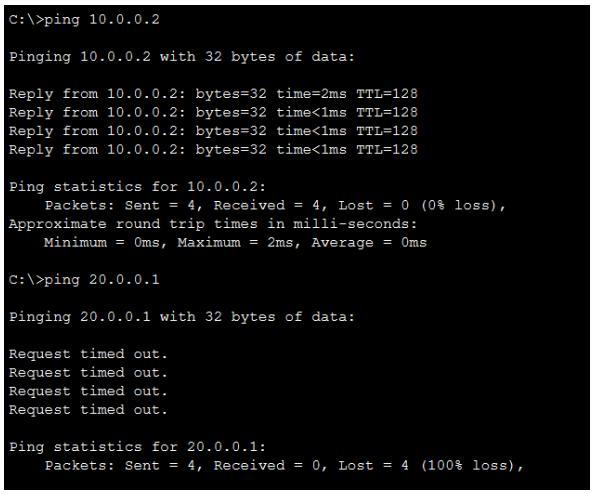
Check the VLANs on the switch to know which ports belong to which VLAN.

```
SW1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   VLAN0010                         active    Fa0/1, Fa0/2
20   VLAN0020                         active    Fa0/3, Fa0/4
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

Step 5:

Test some pings. You should be able to ping between hosts in the same VLAN but not to the other VLAN. You would need a router for this. On 10.0.0.1 host, open CLI and ping other hosts, both in the same VLAN and on different VLANs.



Program 14 : Configuration of VPN using Packet Tracer/ GNS3

Procedure:

Step 1: Create Network Topology with 2 PCS and 3 Routers

Step 2: Configure IP Address for all devices

Device	Interface	IP Address	Default Gateway
PC0	Fa0	192.168.1.2	192.168.1.1
PC1	Fa0	192.168.2.2	192.168.2.1
Router0	Gig0/0	192.168.1.1	-
	Gig0/1	1.1.1.1	-
Router3	Gig0/0	1.1.1.2	-
	Gig0/1	2.2.2.2	-
Router2	Gig0/0	2.2.2.1	-
	Gig0/1	192.168.2.1	-

Step 3: Click on Router 0

Router# config t

Router(config)#interface tunnel 1

Router(config-if)#ip address 50.50.50.1 255.0.0.0

Router(config-if)#tunnel source Gig0/1

Router(config-if)#tunnel destination 2.2.2.1

Router(config-if)no shut

Step 4: Click on Router 2

Router# config t

Router(config)#interface tunnel 1

Router(config-if)#ip address 50.50.50.2 255.0.0.0

Router(config-if)#tunnel source Gig0/0

Router(config-if)#tunnel destination 1.1.1.1

Router(config-if)no shut

Step -5 Router 0

Click on Config - > Click on Static - > Add

Network - 0.0.0.0

Mask - 0.0.0.0

Next Hop - 1.1.1.2

Click on Config - > Click on Static - > Add

Network - 192.168.2.0

Mask - 255.255.255.0

Next Hop – 50.50.50.2

Step -6 Router 2

Click on Config - > Click on Static - > Add

Network - 0.0.0.0

Mask - 0.0.0.0

Next Hop – 2.2.2.2

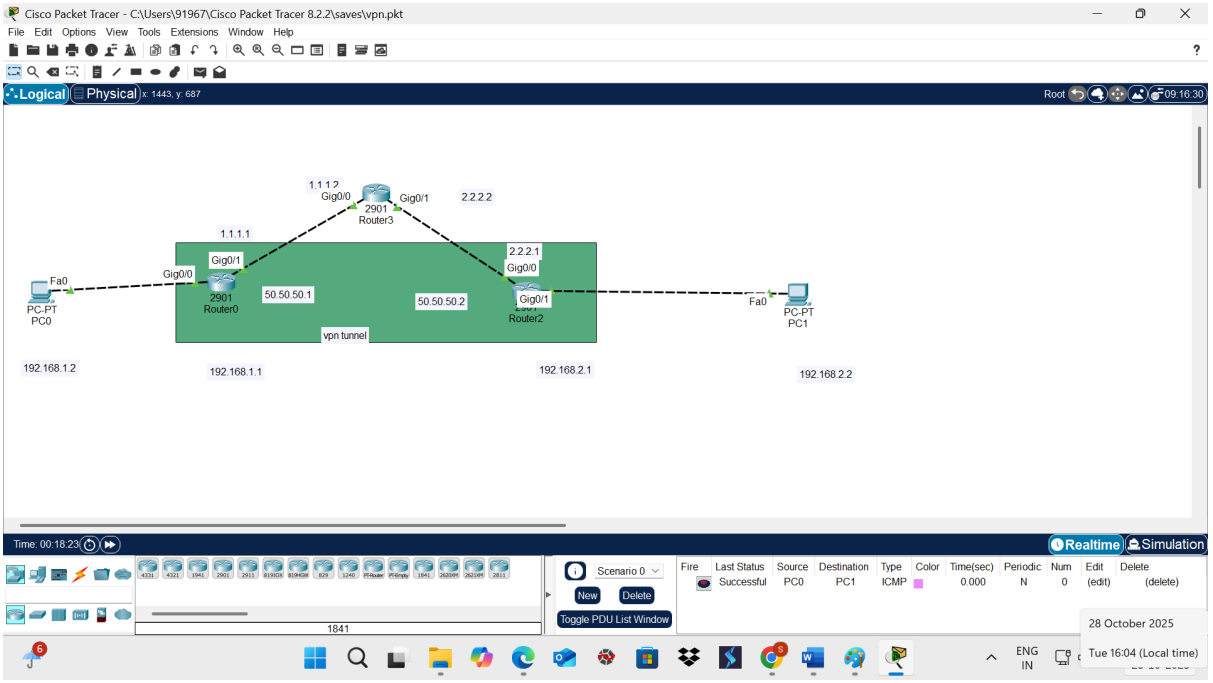
Click on Config - > Click on Static - > Add

Network - 192.168.1.0

Mask - 255.255.255.0

Next Hop – 50.50.50.1

Network Topology



Output

